

SIEM Pipeline

Sysmon and Splunk Enterprise

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Environment: ADForest.local — Windows Server 2025 Domain Controller

Splunk Version: 10.2.0 (Free Licence — 500 MB/day)

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Status: Operational

1. Project Overview

This project implements a fully operational host-based telemetry pipeline on a Windows Server 2025 Active Directory Domain Controller. Microsoft Sysinternals Sysmon captures granular process, file, registry, and network activity. Splunk Enterprise 10.2 ingests, indexes, and surfaces this data through a custom dashboard — replicating the core SIEM workflow used by SOC analysts in enterprise environments.

The build operates within the Splunk free licence limit of 500 MB/day, requiring deliberate event filtering decisions that mirror real-world ingest capacity planning.

Objectives

- Deploy and validate a Sysmon telemetry pipeline from sensor to dashboard
- Configure event filtering to stay within licence ingest constraints
- Resolve sourcetype pipeline conflicts using Splunk best-practice overrides
- Build a reusable process monitoring dashboard with SPL
- Document troubleshooting methodology for portfolio evidence

Skills Demonstrated

Skill Area	Specific Competency
Endpoint Monitoring	Sysmon deployment, config, and operational log management
SIEM Engineering	Splunk input configuration, sourcetype pipeline, index design
Troubleshooting	Log-based diagnosis via splunkd.log and metrics.log
SPL	Process telemetry queries, stats aggregation, dashboard integration
Ingest Management	Event whitelist/blacklist tuning within 500 MB/day constraint
Configuration Management	props.conf override pattern (local/ precedence)

2. Architecture

Splunk Enterprise and Sysmon co-reside on the same Windows Server 2025 host. The Universal Forwarder is not required — Splunk Enterprise's built-in WinEventLog input monitor reads directly from the Sysmon operational event log channel. This is the standard single-node architecture for lab environments.

Component	Version / Detail	Role
Windows Server 2025	Domain Controller (WIN-ESVD1CAD1FJ)	Host OS and AD environment
Sysmon64	Microsoft Sysinternals (latest)	Host telemetry sensor
Splunk Enterprise	10.2.0 (Free Licence)	Indexer, search head, dashboard
Sysmon TA	Community add-on	Field extractions and CIM mapping

Data Flow

- Sysmon64 service captures kernel-level events and writes to: Microsoft-Windows-Sysmon/Operational
- Splunk Enterprise reads the channel via WinEventLog input (renderXml=true)
- Events are indexed into the wineventlog index with sourcetype XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
- The SysmonProcessMonitor dashboard queries the index and surfaces process telemetry

[Screenshot: Architecture diagram / Splunk sourcetype confirmed in index]

3. Configuration

3.1 Sysmon Event Selection

Event IDs were chosen to balance detection coverage against the 500 MB/day ingest ceiling. Network connection events (ID 3) are the primary volume driver on an Active Directory DC and were excluded to keep ingest within limits.

Whitelist — Events Collected

Event ID	Event Name	Detection Value
1	Process Create	Primary process execution telemetry — LOLBins, lateral movement
8	CreateRemoteThread	Process injection detection
11	FileCreate	Malware drop, persistence artefacts
12	Registry Object Added/Deleted	Persistence via registry run keys
13	Registry Value Set	Registry-based persistence and configuration changes
22	DNS Query	C2 beaconing, domain generation algorithms

Blacklist — Events Suppressed

Event ID	Event Name	Reason for Exclusion
3	Network Connect	Extremely high volume on DC — exceeds 500 MB/day limit
10	Process Access	High false positive rate on domain controller workloads

3.2 Splunk inputs.conf

File location: C:\Program Files\Splunk\etc\system\local\inputs.conf

```
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
index      = wineventlog
disabled   = false
current_only = 1
start_from = oldest
renderXml  = true
whitelist  = 1,8,11,12,13,22
blacklist  = 3,10
```

renderXml = true is critical — without it, Splunk ingests events as flat text and the Sysmon TA cannot extract structured fields such as Image, CommandLine, ParentImage, and Hashes.

[Screenshot: inputs.conf open in Notepad showing full Sysmon stanza]

4. Troubleshooting Journey

The pipeline required three distinct issues to be diagnosed and resolved before events appeared in the index. Each is documented below with the diagnostic method and fix applied.

Issue 1 — Universal Forwarder Not Installed

Symptom

SysmonProcessMonitor dashboard returned no results. Search for events under the expected sourcetype returned 0 events.

Diagnostic Steps

- Get-Service SplunkForwarder — returned service not found error
- Attempted to navigate to C:\Program Files\SplunkUniversalForwarder\bin — path did not exist
- Ran index=wineventlog | stats count by sourcetype — returned 0 results, confirming no data in the index

Root Cause

The Universal Forwarder was never installed. In a single-node lab where Splunk Enterprise runs on the monitored host, the forwarder is not required. The inputs.conf in C:\Program Files\Splunk\etc\system\local\ is sufficient for Splunk Enterprise to monitor local Windows Event Logs directly.

Resolution

Confirmed inputs.conf existed at the correct Splunk Enterprise path. Restarted the Splunkd service to force re-reading of the WinEventLog input stanza.

[Screenshot: PowerShell showing SplunkForwarder not found error]

Issue 2 — Events Not Appearing After Restart

Symptom

After Splunkd restart, searches for XmlWinEventLog:Microsoft-Windows-Sysmon/Operational still returned 0 events. Sysmon was confirmed running and generating events via Get-WinEvent.

Diagnostic Steps

- Confirmed Sysmon64 service status: Running
- Ran Get-WinEvent -LogName Microsoft-Windows-Sysmon/Operational -MaxEvents 5 — returned 5 recent Event ID 1 entries, confirming Sysmon was generating events normally
- Examined splunkd.log — found only unrelated errors (quarantine_files.py HTTP 401, clustering licence warnings). No WinEventLog input errors present
- Ran metrics.log query: Select-String wineventlog — confirmed active throughput: series=xmlwineventlog:microsoft-windows-sysmon/operational, ev=27, eps=0.900

Root Cause

The events were ingesting correctly. The apparent 0-result searches were caused by incorrect sourcetype strings in the search bar (XmlEventLog instead of XmlWinEventLog, and a truncated variant from autocomplete). The pipeline was functional throughout.

[Screenshot: PowerShell showing Sysmon64 Running and Get-WinEvent returning Event ID 1 entries]

[Screenshot: metrics.log output confirming xmlwineventlog ingest at 0.9 eps]

Issue 3 — Dashboard Sourcetype Mismatch

Symptom

Direct Splunk searches confirmed events were present, but the SysmonProcessMonitor dashboard continued to show No search results returned.

Diagnostic Steps

- Opened dashboard in Edit mode
- Navigated to Data Sources > Sysmon Process Monitor search > Edit
- Inspected the SPL query — found sourcetype=xm%wineventlog (a truncated/corrupted value, likely from a previous autocomplete selection)

Resolution

Corrected the sourcetype in the dashboard SPL to the full value: XmlWinEventLog:Microsoft-Windows-Sysmon/Operational. Dashboard immediately populated with live process data.

[Screenshot: Dashboard data source editor showing truncated sourcetype and corrected value]

5. Results

Pipeline Validation

Checkpoint	Result	Evidence
Sysmon service running	PASS	Get-Service Sysmon64 → Running
Events in Windows Event Log	PASS	Get-WinEvent → 5 recent Event ID 1 entries
Splunk ingesting events	PASS	metrics.log → 0.9 eps, ev=27
Correct sourcetype in index	PASS	stats count by sourcetype → XmlWinEventLog:Microsoft-Windows-Sysmon/Operational confirmed
Dashboard rendering data	PASS	SysmonProcessMonitor → top 10 processes with counts

Dashboard Output

The SysmonProcessMonitor dashboard displays the top 10 processes by Event ID 1 (Process Create) frequency over the last 24 hours. Post-restart, the table was dominated by Splunk's own internal processes as the index rebuilt — expected and transient behaviour.

Process (Image)	Count	Note
C:\Program Files\Splunk\bin\splunk-optimize.exe	78,814	Splunk index optimisation post-restart
C:\Program Files\Splunk\bin\python3.13.exe	2,978	Splunk internal Python workers
C:\Program Files\Splunk\bin\splunk-powershell.exe	2,950	Splunk PowerShell input
C:\Program Files\Splunk\bin\splunk-MonitorNoHandle.exe	1,475	Splunk monitor process
C:\Program Files\Splunk\bin\splunk-admon.exe	1,475	Splunk AD monitor

[Screenshot: SysmonProcessMonitor dashboard showing populated process table]

Ingest Metrics

Metric	Value
Index	wineventlog
Sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
Events per second (steady state)	~0.9 eps
Active event IDs	1, 8, 11, 12, 13, 22
Suppressed event IDs	3, 10
Licence headroom	Well within 500 MB/day limit

6. Lessons Learned

Technical

- Without XML rendering enabled in `inputs.conf`, Sysmon events arrive as unstructured text. The Sysmon TA cannot parse fields from flat-text events — field extractions silently fail and all SPL field references return null.`renderXml = true`:
- `splunkd.log` captures errors from Splunk's internal processes (Python workers, licensing, telemetry) which are unrelated to input pipeline health. `metrics.log` is the definitive source for verifying whether a specific input is actually ingesting data. Splunk log hierarchy:
- On single-node lab builds where Splunk Enterprise runs on the monitored host, the Universal Forwarder is redundant. Enterprise reads WinEventLog inputs directly. This is a common lab misconception. Universal Forwarder vs Enterprise input:
- Splunk sourcetype strings are case-sensitive and must be specified exactly. A single character difference (`XmlWinEventLog` vs `XmlEventLog`) produces 0 results with no error — the most misleading failure mode in the troubleshooting sequence. Sourcetype precision:

Process

- Confirm the sensor (Sysmon), then the event log (Get-WinEvent), then the ingest (`metrics.log`), then the search (sourcetype string) — before assuming the configuration is wrong. Isolate the pipeline layer by layer:
- Dashboard SPL query fields can silently receive truncated values from browser autocomplete. Always verify the exact string in Edit mode before concluding the data is absent. Autocomplete risk:

7. Next Steps

- Add Suricata alert forwarding from the Topton N150 TAP node into the same Splunk index for cross-layer correlation (network + host)
- Build additional dashboards: failed logon clustering (Event ID 4625), LAPS password retrieval events, BloodHound-flagged path abuse
- Create Splunk alerts for high-confidence indicators: PowerShell with encoded commands (Event ID 1 + CommandLine contains -enc), CreateRemoteThread into lsass.exe (Event ID 8)
- Run a fresh PingCastle healthcheck scan to capture the current hardened score and regenerate PDF evidence for GitHub
- Complete Universal Forwarder deployment on Topton N150 to forward Suricata eve.json alerts to Splunk

Appendix — Full inputs.conf

C:\Program Files\Splunk\etc\system\local\inputs.conf

```
[WinEventLog://Security]
index                = wineventlog
disabled             = false
start_from           = oldest
current_only         = 1
checkpointInterval   = 5
renderXml            = false
blacklist            = 4688,5379,7036,4634

[WinEventLog://System]
index                = wineventlog
disabled             = false
current_only         = 1

[WinEventLog://Application]
index                = wineventlog
disabled             = false
current_only         = 1

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
sourcetype           = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
index                = wineventlog
disabled             = false
current_only         = 1
start_from           = oldest
renderXml            = true
whitelist            = 1,8,11,12,13,22
blacklist            = 3,10
```